

Cross-References in Quarto

Figures, Tables, Equations, and Sections

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2026-06-23

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```
library(survival)
library(ggplot2)
library(kableExtra)
```

1 Study Design

This document demonstrates Quarto’s cross-referencing system. Every figure, table, equation, and section can be referenced by a label — and all references update automatically.

The dataset is described in Section 2. The statistical model is given in Equation 1. Results are shown in Figure 1 and Table 2.

2 Data

The `lung` dataset (NCCTG lung cancer study) contains 228 patients. See Section 1 for context.

3 Statistical Model

The Cox proportional hazards model (Cox 1972) is:

$$h(t \mid \mathbf{x}) = h_0(t) \exp(\mathbf{x}^\top \beta) \quad (1)$$

The concordance index is:

$$C = \frac{\sum_{i:\delta_i=1} \sum_{j:T_j>T_i} \mathbf{1}(\hat{r}_i > \hat{r}_j)}{\sum_{i:\delta_i=1} \sum_{j:T_j>T_i} 1} \quad (2)$$

Equation 1 defines the hazard; Equation 2 defines discrimination.

4 Results

4.1 Descriptive Statistics

Table 1 presents baseline characteristics.

```
data.frame(  
  Variable = c("Age, mean (SD)", "Male sex, n (%)", "Deaths, n (%)"),  
  Value = c(  
    paste0(round(mean(lung$age, na.rm=TRUE), 1),  
            " (", round(sd(lung$age, na.rm=TRUE), 1), ")"),  
    paste0(sum(lung$sex==1, na.rm=TRUE),  
            " (", round(mean(lung$sex==1, na.rm=TRUE)*100, 1), "%)"),  
    paste0(sum(lung$status==2, na.rm=TRUE),  
            " (", round(mean(lung$status==2, na.rm=TRUE)*100, 1), "%)"),  
  )  
) |>  
  kable(booktabs = TRUE) |>  
  kable_styling(latex_options = "hold_position")
```

Table 1: Baseline characteristics of the lung cancer cohort.

Variable	Value
Age, mean (SD)	62.4 (9.1)
Male sex, n (%)	138 (60.5%)
Deaths, n (%)	165 (72.4%)

4.2 Survival Curves

Figure 1 shows overall survival; Figure 2 compares by sex.

```
km <- survfit(Surv(time, status == 2) ~ 1, data = lung)
plot(km, xlab = "Days", ylab = "Survival probability",
     col = "#0072B2", lwd = 2, conf.int = TRUE)
abline(h = 0.5, lty = 2, col = "grey50")
```

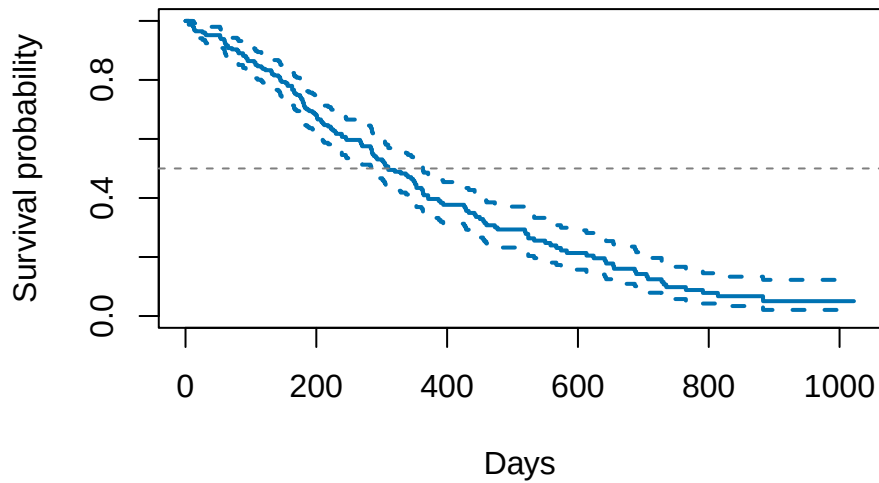


Figure 1: Overall Kaplan-Meier survival curve.

```
km_sex <- survfit(Surv(time, status == 2) ~ sex, data = lung)
plot(km_sex,
     xlab = "Days", ylab = "Survival probability",
     col = c("#D55E00", "#0072B2"), lwd = 2,
     lty = 1:2, conf.int = FALSE)
legend("topright", legend = c("Male", "Female"),
     col = c("#D55E00", "#0072B2"), lwd = 2, lty = 1:2)
```

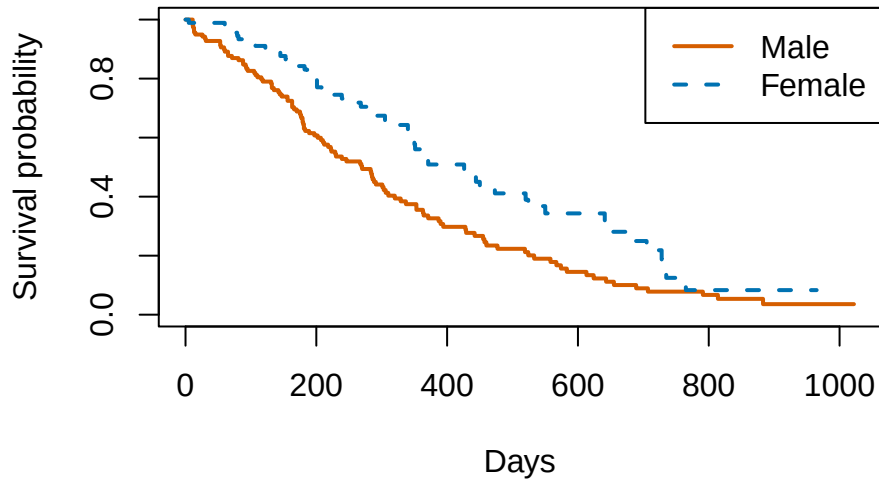


Figure 2: Kaplan-Meier curves stratified by sex.

4.3 Cox Regression

Results from fitting Equation 1 are in Table 2.

```
fit <- coxph(Surv(time, status==2) ~ age + sex + ph.ecog, data=lung)

broom::tidy(fit, exponentiate = TRUE, conf.int = TRUE) |>
  dplyr::select(term, estimate, conf.low, conf.high, p.value) |>
  dplyr::mutate(dplyr::across(where(is.numeric), \(x) round(x, 3))) |>
  kable(booktabs = TRUE,
        col.names = c("Variable", "HR", "95% CI Low",
                      "95% CI High", "p-value")) |>
  kable_styling(latex_options = "hold_position")
```

Table 2: Cox model: hazard ratios with 95% confidence intervals.

Variable	HR	95% CI Low	95% CI High	p-value
age	1.011	0.993	1.030	0.232
sex	0.575	0.414	0.799	0.001
ph.ecog	1.590	1.273	1.986	0.000

5 Summary

Cross-references in Quarto:

Object	Label prefix	Reference
Figure	<code>fig-</code>	<code>@fig-km</code>
Table	<code>tbl-</code>	<code>@tbl-cox</code>
Equation	<code>eq-</code>	<code>@eq-cox</code>
Section	<code>sec-</code>	<code>@sec-methods</code>

All labels update automatically when objects are added or reordered.

References

Cox, D. R. 1972. “Regression Models and Life-Tables.” *Journal of the Royal Statistical Society: Series B* 34 (2): 187–220. <https://doi.org/10.1111/j.2517-6161.1972.tb00899.x>.